

LOUIS ALMAIRAC

Master MVA student at ENS Paris-Saclay and engineering student at École des Ponts,
working on machine learning for time-series and biosignals, and building products end-to-end.

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EDUCATION

École Normale Supérieure Paris-Saclay • Paris, France 2026 – 2027

Master MVA (Mathematics, Vision, Learning): machine learning, computer vision, statistics and signal processing.

École Nationale des Ponts et Chaussées • Paris, France 2022 – 2027

Master of Engineering: Mathematics and Computer Science

Relevant courses: Machine Learning, Stochastic Processes, Deep Learning, Statistics, Computer Vision, Programming

Lycée Pierre de Fermat • Toulouse, France 2020 – 2022

Intensive Mathematics, Physics and Computer Science courses preparing for France's top engineering schools.

EXPERIENCES

Research Intern, Université du Québec à Montréal • Montréal, Canada March 2026 – July 2026

- Classified flow states from EEG at MicroPro Lab (Pr. Mounir Boukadoum), combining empirical mode decomposition, time-frequency features and Riemannian geometry on covariance matrices.
- Identified a labelling confound in the reference dataset, which caps how well flow can be separated within gameplay; manuscript in preparation for IEEE ISCAS.

Machine Learning Intern, Caisse des Dépôts • Paris, France June 2025 – Dec. 2025

- Implemented a LambdaMART learning-to-rank model selecting countries within an emerging-market equity universe, replacing a neural ListMLE baseline and optimising NDCG directly.
- Engineered the financial features (momentum, volatility, FX-equity correlation, drawdowns) and a purged, embargoed temporal cross-validation ruling out look-ahead bias.

Medical Time-Series Research Project, Ponts et Chaussées • Paris, France Nov. 2024 – June 2025

- Developed a time-frequency Granger causality framework combining wavelet decomposition and VAR modelling to analyse EEG time-series during sleep, highlighting state-dependent brain interactions.
- Presented the work at the International Summer Institute in Network Physiology (Frontiers, Como, Italy, 2025).

Data Scientist Intern, Sopra Steria • Paris, France Jan. 2024 – June 2024

- Cleaned inconsistent operational data and built the analysis tools identifying the causes of metro traffic delays, for an RATP line-automation project.

PROJECTS & COMPETITIONS

Martha, AI Stylist App (personal project) • Paris, France Currently

Designing and shipping a native iOS app (SwiftUI, CoreData, Supabase) that digitises a wardrobe and recommends outfits with LLMs and vision models, from data model and backend to interface.

Medical Data Challenge, QRT • Paris, France April 2025

Built survival prediction models for blood cancer patients from clinical and molecular data, evaluated with IPCW-C-index, in collaboration with Institut Gustave Roussy.

Hackathons, Mistral AI x Alan & Nuit pour Entreprendre • Paris, France 2024

Ranked 4th out of 100+ participants with an LLM-powered food-waste prevention app built in 24h; won the Grand Prix of Nuit pour Entreprendre with Blue Seed, an AI irrigation-planning solution for farmers pitched in 10h.

Mathematics Competition, TFJM² • Toulouse and Paris, France 2020 and 2023

Achieved 2nd place nationally as part of an 8-member team; Community Manager of the 2023 edition.

SKILLS & INTERESTS

Languages French (native), English (fluent), Spanish (fluent)

Programming Advanced skills in Python (PyTorch, scikit-learn, Pandas), C++, Swift (SwiftUI)

Leadership Secretary General of Forum Trium, a 4-school student-business forum (EUR 400k budget)

Hobbies Competitive Rugby player, Member of English Debating Club, Hackathons